Matrices - 1

→ Rectangular Arrangement of numbers is a matrix.

$$a_{ij} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3\times 3}$$

← Rows; ↑ Columns; 3×3 → Order of matrix

i = No. of Rows, j = No. of columns

i.e a_{23} → Element in 2nd Row and 3rd column of the matrix.

→ No. of elements in matrix = Multiplication of order/rows & columns.

Eg: A matrix has 8 elements. What can be its possible orders?

Sol^n orders = 8×1, 1×8, 4×2, 2×4] Both are diff.] Both are diff.

Types of Matrices:

1) Row Matrix ⇒ Has only 1 Row.
$A = [1 \;\; 4 \;\; 8 \;\; 7]_{1\times4}$

2) Column Matrix ⇒ Has only 1 column
$A = \begin{bmatrix} 3 \\ 5 \\ 4 \end{bmatrix}_{3\times1}$

3) Null/zero Matrix (O) ⇒ All elements are zero
$A = O = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{2\times3}$

4) Horizontal Matrix ⇒ Rows < Columns
Vertical Matrix ⇒ Rows > Columns
Square Matrix ⇒ Rows = Columns

Square Matrix:

* Pair of elements (a_{ij}) & (a_{ji}) ⇒ (Conjugate Elements)
* Elements $a_{11}, a_{22} \dots a_{nn}$ ⇒ (Diagonal Elements)
* Line along which diagonal elements lie is called 'Principal/Leading Diagonal'

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3\times3}$$

Trace(A) = Tr(A) = Sum of all Diagonal elements

$Tr(A) = a_{11} + a_{22} + a_{33}$

Types of Square Matrix:→

Square Matrix
- Triangular Matrix
  - Upper Triang. Matrix: $\begin{bmatrix} x & x & x \\ 0 & x & x \\ 0 & 0 & x \end{bmatrix}$ Down ∆, all zero
  - Lower Triang. Matrix: $\begin{bmatrix} x & 0 & 0 \\ x & x & 0 \\ x & x & x \end{bmatrix}$ Up ∆, all zero
- Diagonal Matrix: Except diagonal elements all are zero.
  i.e $a_{ij} = 0$ if $i \neq j$
  $A = \begin{bmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{bmatrix}_{3\times3}$

* Diagonal matrix is abbreviated as
$A = dia(d_1, d_2, d_3)$

Scalar Matrix: All diagonal elements have same value
i.e $d_1 = d_2 = d_3 \dots = d_n = a$
$\begin{bmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{bmatrix}$

Unit Matrix: All diagonal elements are 1.
i.e $d_1 = d_2 \dots = d_n = 1$
$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Algebra of Matrices:

Addition / Substraction :→

We can add or substract two matrices only if they are of same order.

Eg: $A = \begin{bmatrix} \sqrt{5} & 1 & -1 \\ 2 & 3 & 0 \end{bmatrix}$ & $B = \begin{bmatrix} 2 & \sqrt{3} & 1 \\ -2 & 3 & 1/2 \end{bmatrix}$

$$A+B = \begin{bmatrix} \sqrt{5}+2 & 1+\sqrt{3} & -1+1 \\ 2-2 & 3+3 & 0+1/2 \end{bmatrix} = \begin{bmatrix} \sqrt{7}+\sqrt{5} & 0 \\ 0 & 6 & 1/2 \end{bmatrix}$$

$$A-B = \begin{bmatrix} \sqrt{5}-2 & 1-\sqrt{3} & -1-1 \\ 2+2 & 3-3 & 0-1/2 \end{bmatrix} = \begin{bmatrix} \sqrt{3}-\sqrt{5} & -2 \\ 4 & 0 & -1/2 \end{bmatrix}$$

Properties of addn & substraction ⇒
1) $A+B = B+A$ (Commutative)
2) $(A+B)+C = A+(B+C)$ (Associative)
3) $Tr(A \pm B) = Tr(A) \pm Tr(B)$
4) $Tr(KA) = K \cdot Tr(A)$

Multiplication of Matrix by a Scalar :→

→ Scalar is multiplied by all the elements of a matrix.

$$KA = \begin{bmatrix} Ka_{11} & Ka_{12} & Ka_{13} \\ Ka_{21} & Ka_{22} & Ka_{23} \\ Ka_{31} & Ka_{32} & Ka_{33} \end{bmatrix}$$

* $K(A+B) = KA + KB$

Additive Inverse ⇒ Jis matrix ko add karne pe null matrix aa jaye.
i.e $A+B = O = B+A$
(A & B matrix are additive inverse of each other)

Multiplication of Matrices :→

→ Matrix multiplication is possible only if
(No. of columns of first matrix) = (No. of rows of second matrix)

i.e $A_{m\times n} \cdot B_{p\times q} = C_{m\times q}$
Only if $(n = p)$

→ Matrix multiplication is always Row by Column.

Eg: $\begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 2 \end{bmatrix} = A$

Sol^n

	(1,2)	(2,5)	(3,2)
(2,1)	2+2	4+5	6+2
(3,5)	3+10	6+25	9+10

$$\Rightarrow A = \begin{bmatrix} 4 & 9 & 8 \\ 13 & 31 & 19 \end{bmatrix}$$ Ans

Properties of Matrix multiplication ⇒

* It is non-commutative. i.e $AB \neq BA$
* Multiplication with identity matrix as gives the same matrix.
  i.e $I \cdot A = A \cdot I = A$

Imp
* If $A = O$ or $B = O \Rightarrow AB = 0$
  But, if $AB = O \not\Rightarrow A = O$ or $B = O$
  Matrix multiplication for non-singular matrices can also be zero.

* If $AB = AC \not\Rightarrow B = C$
  But if $B = C \Rightarrow AB = AC \Rightarrow BA = CA$

* In case if,
  $AB = BA \Rightarrow$ A & B commute each other
  $AB = -BA \Rightarrow$ A & B anti-commute each other

* Multiplication of diagonal matrices of same order is always commutative.

Eg: $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

Then $AB = BA = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 6 \end{bmatrix}$

We can directly multiply diag. elements of diag. matrices during matrix multiplication.

* Follows Associative & Distributive properties.
  i.e $(A \cdot B) \cdot C = A \cdot (B \cdot C)$
  $A(B+C) = AB + AC$
  $(B+C)A = BA + CA$] Position matters here

Positive Integral powers of a Sq. matrix :→

1) $A^2 \cdot A = (A \cdot A)A = A(A \cdot A) = A^3$
2) $I^m = I \;\; (m \in N)$
3) $A^m \cdot A^n = A^{m+n}$ and $(A^m)^n = A^{mn}$
4) $A^0 = I_n$ (n is same as order of A)

Ques Let $A=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B=\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ such that $AB=B$ and $a+d=2021$, then find $ad-bc$.

Sol^n It is given, $a+d=2021$

Here, we can assume any one constant as zero. So, $[d=0]$

It is given, $AB=B$

$$\begin{bmatrix} 2021 & b \\ c & 0 \end{bmatrix}\begin{bmatrix} \alpha \\ \beta \end{bmatrix}=\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

$$\begin{bmatrix} 2021\alpha+b\beta \\ c\alpha \end{bmatrix}=\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

$2021\alpha+b\beta=\alpha$ $\quad [\beta=c\alpha]$

$2021\alpha+b(c\alpha)=\alpha$ $\quad [\because ad=a(0)=0]$

$bc=-2020 \Rightarrow \boxed{-bc=2020}$ Ans

Ques If A is a 2x2 matrix such that trace of A is 3 and trace of A^3 is -18, then find $|A|$.

Sol^n Let, $A=\begin{bmatrix} 3 & b \\ c & 0 \end{bmatrix}$

$$A^3=\begin{bmatrix} 3 & b \\ c & 0 \end{bmatrix}\begin{bmatrix} 3 & b \\ c & 0 \end{bmatrix}\begin{bmatrix} 3 & b \\ c & 0 \end{bmatrix}$$

$$=\begin{bmatrix} 9+bc & 3b \\ 3c & bc \end{bmatrix}\begin{bmatrix} 3 & b \\ c & 0 \end{bmatrix}=\begin{bmatrix} (9+bc)(3)+3bc & — \\ — & 3bc \end{bmatrix}$$

$Tr(A^3)=27+3bc+3bc+3bc=-18$

$3+bc=-2 \Rightarrow \boxed{bc=-5}$

$|A|=3(0)-bc \Rightarrow \boxed{-bc=5}$ Ans

Ques Let $A=\begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}$ and $A^4=[a_{ij}]$. If $a_{11}=109$, find a_{22}.

Sol^n

$$A^2=\begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}=\begin{bmatrix} x^2+1 & x \\ x & 1 \end{bmatrix}$$

$$A^4=A^2\cdot A^2=\begin{bmatrix} x^2+1 & x \\ x & 1 \end{bmatrix}\begin{bmatrix} x^2+1 & x \\ x & 1 \end{bmatrix}$$

$$A^4=\begin{bmatrix} (x^2+1)^2+x^2 & — \\ — & x^2+1 \end{bmatrix}$$

$a_{11}=(x^2+1)^2+x^2=109$

$x^4+1+3x^2=109$

$x^4+3x^2-108=0$

$(x^2+12)(x^2-9)=0$ $\quad x^2=-12$ ✗, $x^2=9$ ✓

So, $a_{22}=x^2+1=9+1$

$\boxed{a_{22}=10}$ Ans

Ques Let, $A=\begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$ such that $A^{32}=\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Then, a value of α is ____.

Sol^n

$$A^{32}=\begin{bmatrix} \cos 32\alpha & -\sin 32\alpha \\ \sin 32\alpha & \cos 32\alpha \end{bmatrix}=\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$\cos 32\alpha=0 \Rightarrow 32\alpha=\pi/2 \Rightarrow \boxed{\alpha=\pi/64}$ Ans

Ques If $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\cdot\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}\cdots\begin{bmatrix} 1 & n-1 \\ 0 & 1 \end{bmatrix}=\begin{bmatrix} 1 & 78 \\ 0 & 1 \end{bmatrix}$ then value of n is ____.

Sol^n On successive matrix multiplications, the a_{12} elements are getting added.

i.e $\begin{bmatrix} 1 & 1+2+3\cdots+(n-1) \\ 0 & 1 \end{bmatrix}=\begin{bmatrix} 1 & 78 \\ 0 & 1 \end{bmatrix}$

$\frac{n(n-1)}{2}=78 \Rightarrow \boxed{n=13}$ Ans

Important Matrices with Pattern

1) In case of rotation matrix, the angle in multiplied by the power to which matrix is raised.

$A=\begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$ | $A^n=\begin{bmatrix} \cos(n\alpha) & -\sin(n\alpha) \\ \sin(n\alpha) & \cos(n\alpha) \end{bmatrix}$

2) In case of following pattern, simply add a_{12} terms on successive multiplication.

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}\cdots\begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}=\begin{bmatrix} 1 & 1+2+3\cdots+n \\ 0 & 1 \end{bmatrix}$$

Ques If $A=\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}$ then find $A^{2025}-A^{2020}$

Sol^n

$$A^2=\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}=\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$A^3=\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}\cdot\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}=\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

i.e $A^n=\begin{bmatrix} 1 & 0 & 0 \\ n-1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

$$A^{2025}-A^{2020}=\begin{bmatrix} 0 & 0 & 0 \\ 2024-2019 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{A^{2025}-A^{2020}=\begin{bmatrix} 0 & 0 & 0 \\ 5 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}$$ Ans

Ques If $P=\begin{bmatrix} 1 & 0 \\ 1/2 & 1 \end{bmatrix}$ then find P^{50}.

Sol^n

$$P^2=\begin{bmatrix} 1 & 0 \\ 1/2 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 1/2 & 1 \end{bmatrix}=\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

$$P^3=\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 1/2 & 1 \end{bmatrix}=\begin{bmatrix} 1 & 0 \\ 3/2 & 1 \end{bmatrix}$$

i.e $P^n=\begin{bmatrix} 1 & 0 \\ n/2 & 1 \end{bmatrix} \Rightarrow \boxed{P^{50}=\begin{bmatrix} 1 & 0 \\ 25 & 1 \end{bmatrix}}$ Ans

Ques Let M be a 3x3 matrix satisfying $M\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}=\begin{bmatrix} -1 \\ 3 \\ 3 \end{bmatrix}$, $M\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}=\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ and $M\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 12 \end{bmatrix}$. Then, the sum of diagonal entries of M is ____.

Sol^n

$$\begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}=\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

$a_1-a_2=1 \Rightarrow \boxed{a_1=0}$

$b_1-b_2=1 \Rightarrow \boxed{b_1=3}$

$c_1-c_2=-1 \Rightarrow \boxed{c_1=2}$

$$\begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}=\begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix} \Rightarrow 0+a_2+0=-1$$

$\Rightarrow \boxed{a_2=-1}$

$\boxed{b_2=2}$ $\quad \boxed{c_2=3}$

$$\begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 12 \end{bmatrix} \Rightarrow c_1+c_2+c_3=12$$

$\boxed{c_3=7}$

$Tr(M)=a_1+b_2+c_3 \Rightarrow 0+2+7$

$\boxed{Tr(M)=9}$ Ans

Transpose of a Matrix

→ Interchange rows and columns of a matrix to take its transpose.

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}^T=\begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$$

Properties of Transpose :→

1) $(A^T)^T=A$
2) $(A\pm B)^T=A^T\pm B^T$
3) $(kA)^T=k(A^T)$
4) $(AB)^T=B^TA^T$ (Reversal Property)

$(ABC)^T=C^TB^TA^T$

Ques Let $A+2B=\begin{bmatrix} 1 & 2 & 0 \\ 6 & -3 & 3 \\ -5 & 3 & 1 \end{bmatrix}$ and $2A-B=\begin{bmatrix} 2 & -1 & 5 \\ 2 & -1 & 6 \\ 0 & 1 & 2 \end{bmatrix}$. Then, $Tr(A)-Tr(B)$ is equal to ____.

Sol^n Since we need to find Traces, simply take trace of the given equations.

$Tr(A+2B)=Tr\begin{bmatrix} 1 & 2 & 0 \\ 6 & -3 & 3 \\ -5 & 3 & 1 \end{bmatrix} \Rightarrow Tr(A)+2Tr(B)=-1$ —①

$Tr(2A-B)=Tr\begin{bmatrix} 1 & -1 & 5 \\ 2 & -1 & 6 \\ 0 & 1 & 2 \end{bmatrix} \Rightarrow 2Tr(A)-Tr(B)=3$ —②

②×2 + ① ⇒

$Tr(A)+2Tr(B)=-1$

$4Tr(A)-2Tr(B)=6$

$5Tr(A)=5 \Rightarrow \boxed{Tr(A)=1}$ $\quad \boxed{Tr(B)=-1}$

So, $\boxed{Tr(A)-Tr(B)=2}$ Ans

Ques If for matrix $A=\begin{bmatrix} 1 & -\alpha \\ \alpha & \beta \end{bmatrix}$, $A\cdot A^T=I_2$, then find value of $\alpha^4+\beta^4$.

Sol^n $\because A\cdot A^T=I_2 \Rightarrow \begin{bmatrix} 1 & -\alpha \\ \alpha & \beta \end{bmatrix}\begin{bmatrix} 1 & \alpha \\ -\alpha & \beta \end{bmatrix}=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$\Rightarrow \begin{bmatrix} 1+\alpha^2 & \alpha-\alpha\beta \\ \alpha-\alpha\beta & \alpha^2+\beta^2 \end{bmatrix}=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$1+\alpha^2=1 \Rightarrow \boxed{\alpha=0}$ | $\alpha^2+\beta^2=1$ | $\boxed{\beta^2=1}$

Hence, $\boxed{\alpha^4+\beta^4=1}$ Ans

Ques If P is a 3x3 matrix such that $P^T=2P+I$ then find its relation with $X=\begin{bmatrix} x \\ y \\ z \end{bmatrix}\neq\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

Sol^n $P^T=2P+I$

$(P^T)^T=(2P+I)^T \Rightarrow P=2P^T+I$

$P=2(2P+I)+I \Rightarrow P=4P+3I$

$O=3P+3I \Rightarrow \boxed{P=-I}$

⇓ Multiply by X

$PX=-IX \Rightarrow \boxed{PX=-X}$ Ans

Matrices - 2

Ques If $P = \begin{bmatrix} \sqrt{3}/2 & 1/2 \\ -1/2 & \sqrt{3}/2 \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Q = PAP^T$ and $X = P^T Q^{2005} P$, then X is equal to ____.

Sol^n $X = P^T Q \cdot Q \cdot Q \cdots Q \cdot P$

$X = P^T PAP^T \cdot PAP^T \cdot PAP^T \cdots PAP^T \cdot P$

$\left(P \cdot P^T = \begin{bmatrix} \sqrt{3}/2 & 1/2 \\ -1/2 & \sqrt{3}/2 \end{bmatrix} \cdot \begin{bmatrix} \sqrt{3}/2 & -1/2 \\ 1/2 & \sqrt{3}/2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right)$

$X = I \cdot A \cdot I \cdot A \cdot I \cdot A \cdots I \cdot A \cdot I$

$X = A^{2005}$ (We know, $A^n = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$)

$X = \begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$ Ans

Symmetric and Skew Symmetric Matrix :

(This classification is only for square matrices)

Symmetric Matrix

* $A^T = A$ is necessary condition
* All elements about principal diagonal are mirror images.

Eg $\begin{bmatrix} \sqrt{3} & 2 & 3 \\ 2 & -1.5 & -1 \\ 3 & -1 & 1 \end{bmatrix}$

$a_{ij} = a_{ji}$

* For any square matrix A:
  - $A + A^T$ ⇒ Always symmetric
  - $A - A^T$ ⇒ Always skew symmetric

Skew Symmetric Matrix

* $A^T = -A$ is necessary cond^n
* All elements of principal diagonal are zero.
* Elements about principal diagonal are same in magnitude but opp^n sign.

Eg: $\begin{bmatrix} 0 & 2 & 4 \\ -2 & 0 & 7 \\ -4 & -7 & 0 \end{bmatrix}$

$a_{ij} = -a_{ji}$

Imp
* Determinant value of odd-order skew symm. matrix is always zero(0).

i.e $\begin{vmatrix} 0 & 2 & 4 \\ -2 & 0 & 7 \\ -4 & -7 & 0 \end{vmatrix}_{3\times3} = 0$

NOTE: Thus any square matrix can be represented as the sum of a symmetric & skew symmetric matrix.

$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$$

(symmetric) (skew-symmetric)

Ques Let A be a symm. matrix of order 2 with integer entries. If the sum of the diagonal elements of A^2 is 1, then the possible number of such matrices is ____.

Sol^n Let the matrix be,

$A = \begin{bmatrix} a & c \\ c & b \end{bmatrix}$ and $a, b, c \in Z$

Now, $A^2 = \begin{bmatrix} a & c \\ c & b \end{bmatrix} \begin{bmatrix} a & c \\ c & b \end{bmatrix} = \begin{bmatrix} a^2+c^2 & ___ \\ ___ & c^2+b^2 \end{bmatrix}$

$Tr(A^2) = 1 \Rightarrow a^2 + b^2 + 2c^2 = 1$

Means 4 such matrices possible. ⇐ $\begin{matrix} (1)^2 & 0 \\ (-1)^2 & 0 \end{matrix}$ | $\begin{matrix} 0 & 0 \\ 0 & 0 \end{matrix}$ | $\begin{matrix} (1)^2 & 0 \\ 0 & (1)^2 \end{matrix}$ | $\begin{matrix} 0 & 0 \\ 0 & 0 \end{matrix}$ | $\begin{matrix} 0 & 0 \\ (-1)^2 & 0 \end{matrix}$ | $\begin{matrix} 0 & 0 \\ 0 & (-1)^2 \end{matrix}$ — a, b & c can have only these set of values

Properties of Symmetric and skew symm. matrix :⇒

1) If A is a symmetric matrix then

$-A$, A^n, KA, A^{-1}, A^T, B^TAB — These all matrices are also symmetric

(B = sq. matrix of same order as A)

2) If A is a skew-symmetric matrix, then

A^{2n+1}, KA, B^TAB — skew symmetric Matrix

A^{2n} ⇒ symmetric Matrix

3) If A & B are two symmetric matrices, then

$A \pm B$, $AB + BA$ ⇒ Always symmetric

AB ⇒ Symmetric only when $AB = BA$

$AB - BA$ ⇒ Always skew symmetric

4) If A & B are two skew-symmetric matrix, then

$A \pm B$, $AB - BA$ ⇒ skew symmetric

$AB + BA$ ⇒ Symmetric matrix.

Ques Let X and Y be two 3x3 skew-symm. matrices and Z be a 3x3 symmetric matrix. Find which of the following are skew symm.?

1) $Y^3Z^4 - Z^4Y^3$ $[X^T = -X]$, $[Y^T = -Y]$, $[Z^T = Z]$

Sol^n $P^T = (Y^3Z^4 - Z^4Y^3)^T \Rightarrow (Y^3Z^4)^T - (Z^4Y^3)^T$

$= (Z^T)^4 (Y^T)^3 - (Y^T)^3 (Z^T)^4$

$= -Z^4Y^3 + Y^3Z^4 = P$

i.e $P = Y^3Z^4 - Z^4Y^3$ is symm. & not skew symmetric.

2) $X^{44} + Y^{44}$

Sol^n $P^T = (X^{44} + Y^{44})^T = (X^{44})^T + (Y^T)^{44}$

$= X^{44} + Y^{44} = P$

i.e It is not skew symmetric.

3) $X^{23} + Y^{23}$

Sol^n $P^T = (X^{23} + Y^{23})^T = (X^T)^{23} + (Y^T)^{23}$

$= -X^{23} - Y^{23} = -(X^{23} + Y^{23})$

$= -P$

i.e $P = X^{23} + Y^{23}$ is a skew symmetric matrix

4) $X^4Z^3 - Z^3X^4$

Sol^n $P^T = (X^4Z^3 - Z^3X^4)^T = (X^4Z^3)^T - (Z^3X^4)^T$

$= (Z^T)^3 (X^T)^4 - (X^T)^4 (Z^T)^3$

$= Z^3X^4 - X^4Z^3 = -P$

i.e $P = X^4Z^3 - Z^3X^4$ is a skew symmetric matrix.

Types of Matrices :

1) Orthogonal Matrix ⇒ $A^TA = AA^T = I$
2) Idempotent Matrix ⇒ $A^2 = A$
3) Involutory Matrix ⇒ $A^2 = I_n$
4) Nilpotent Matrix ⇒ $A^K = O$ (K ⇒ index of nilpotency)
5) Periodic Matrix ⇒ $A^{K+1} = A$ (K = Period)

NOTE: Important Concept ⇒

If, $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$, then

$Tr(AA^T) = a_{11}^2 + a_{12}^2 + a_{13}^2 + a_{21}^2 + a_{22}^2 + a_{23}^2 + a_{31}^2 + a_{32}^2 + a_{33}^2$

(Sum of square of all the elements)

Ques How many 3x3 matrices M with entries $\{0, 1, 2\}$ are possible for which the sum of diagonal entries of M^TM is 5?

Sol^n $Tr(M^TM) = \sum_{i,j=1}^{3,3} a_{ij}^2$

$= a_{11}^2 + a_{12}^2 + a_{13}^2 + a_{21}^2 + a_{22}^2 + a_{23}^2 + a_{31}^2 + a_{32}^2 + a_{33}^2$

$= 1+1+1+1+1+0+0+0+0$

$= 4+1+0+0+0+0+0+0+0$

$\frac{9!}{7!}$ $\quad$ $\frac{9!}{5!4!}$

Total matrices $M = \frac{9!}{7!} + \frac{9!}{5!4!} = 198$ Ans

Ques The maximum no. of 3x3 M matrices with entries $\{0, 1, 2\}$ possible for which $Tr(M^TM) = 7$ is ____.

Sol^n $Tr(M^TM) = 1+1+1+1+1+1+1+0+0$

$= 4+1+1+1+0+0+0+0+0$

No. of $M = \frac{9!}{7!2!} + \frac{9!}{3!5!} = 36 + 504$

$= 540$ Ans

Adjoint of a Matrix A :

$$\text{Adjoint} = (\text{Cofactor Matrix})^T$$

$\text{Adjoint} = \begin{bmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{bmatrix}$

Eg: Find the adjoint of $\begin{bmatrix} 3 & 1 & -1 \\ 2 & -2 & 0 \\ 1 & 2 & -1 \end{bmatrix}$

Sol^n $M = \begin{bmatrix} 2 & -2 & 6 \\ 1 & -2 & 5 \\ -2 & 2 & -8 \end{bmatrix} \Rightarrow C = \begin{bmatrix} 2 & 2 & 6 \\ -1 & -2 & -5 \\ -2 & -2 & -8 \end{bmatrix}$

$\text{Adjoint} = (C)^T = \begin{bmatrix} 2 & -1 & -2 \\ 2 & -2 & -2 \\ 6 & -5 & -8 \end{bmatrix}$ Ans

Properties of Adjoint :⇒

1) $A(\text{adj}\,A) = (\text{adj}\,A)A = |A| I_n$
2) $\text{adj}(AB) = (\text{adj}\,B) \cdot (\text{adj}\,A)$
3) $\text{adj}(KA) = K^{n-1} (\text{adj}\,A)$
4) $\text{adj}(\text{adj}\,A) = |A|^{n-2} \cdot A$

5) $|adjA| = |A|^{n-1}$

$|adj(adjA)| = |A|^{(n-1)^2}$ ← Jitne adj honge, utni (n-1) ki power hogi

6) $\boxed{adj(A^{-1}) = (adjA)^{-1} = \frac{A}{|A|}}$

Ques If the matrices $A=\begin{bmatrix}1&1&2\\1&3&4\\1&-1&3\end{bmatrix}$, $B=adjA$ and $C=3A$, then $\frac{|adjB|}{|C|}$ is equal to ____.

Sol^n

$$\frac{|adjB|}{|C|} = \frac{|adj\,adjA|}{|3A|} = \frac{|A|^{2^2}}{3^3|A|} = \frac{|A|^3}{27}$$

$$= \frac{6^3}{3^3} \Rightarrow 2^3 = 8 \quad \text{Ans}$$

Ques Let A be a 3×3 matrix. If $det(2adj(2adj(adj(2A))))) = 2^{41}$, then value of $det(A^2)$ is equal to ____.

Sol^n

$$2^3 \,|\, adj(2adj(adj2A))| = 2^{41}$$

$$2^3\,|2adj(adj2A)|^2 = 2^{41} \quad (\because |adj\square| = |\square|^{n-1})$$

$$2^3\,(2^3)^2\,|adj(adj2A)|^2 = 2^{41}$$

$$2^9\,|adj2A|^4 = 2^{41} \Rightarrow 2^9\,|2A|^8 = 2^{41}$$

$$2^9\,(2^3)^8\,|A|^8 = 2^{41} \Rightarrow |A|^8 = 2^8$$

$$\Rightarrow \boxed{|A| = \pm 2}$$

So, $\boxed{|A^2| = 4}$ Ans

Ques Let K be a +ve real number and let

$A=\begin{bmatrix}2K-1 & 2\sqrt{K} & 2\sqrt{K}\\ 2\sqrt{K} & 1 & -2K\\ -2\sqrt{K} & 2K & 1\end{bmatrix}$ and $B=\begin{bmatrix}0 & 2K-1 & \sqrt{K}\\ 1-2\sqrt{K} & 0 & 2\sqrt{K}\\ -\sqrt{K} & -2\sqrt{K} & 0\end{bmatrix}$

If $|adjA| + |adjB| = 10^6$, then $[K]$ is ____.

Sol^n

$$|adjA| + |adjB| = 10^6$$

$$|A|^2 + |B|^2 = 10^6 \Rightarrow \boxed{|A| = 10^3}$$

(∵ Skew symm matrix of odd order)

Now,

$$|A| = (2K+1)^3 = 10^3 \Rightarrow 2K+1 = 10$$

$$\Rightarrow K = \frac{9}{2} \Rightarrow \boxed{[K] = 4} \quad \text{Ans}$$

Inverse of a Matrix :-

A square matrix 'A' is said to be invertible (non-singular) if there exists a matrix B such that,

$$\boxed{AB = I = BA}$$

'B' is called inverse of 'A' & denoted as $\boxed{A^{-1}=B}$

* A^{-1} can be calculated for any sq. matrix as,

$$\boxed{A^{-1} = \frac{(adjA)}{|A|}}$$

↳ A^{-1} exist only for the non-singular matrices A i.e. $(|A| \neq 0)$

Shortcut for Inverse of 2×2

If $A=\begin{bmatrix}a&b\\c&d\end{bmatrix}$ then its adjoint can be written directly by

(i) Exchange the positions of a & d

(ii) Reverse the signs of b & c

i.e. $\boxed{A^{-1} = \frac{1}{|A|}\begin{bmatrix}d&-b\\-c&a\end{bmatrix}}$

Eg: $\begin{bmatrix}4&7\\2&6\end{bmatrix} \Rightarrow A^{-1} = \frac{1}{10}\begin{bmatrix}6&-7\\-2&4\end{bmatrix}$

Shortcut for Inverse of Diagonal Matrices

Simply inverse the non zero elements

i.e. $\begin{bmatrix}1&0&0\\0&2&0\\0&0&3\end{bmatrix}^{-1} = \begin{bmatrix}1&0&0\\0&1/2&0\\0&0&1/3\end{bmatrix}$

Shortcut for Inverse of a 3×3 Matrix

We can directly write the adjoint matrix by following steps

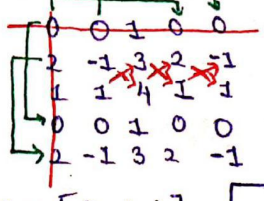
(i) Rewrite the first two row and columns.

(ii) Now, delete First Row & First column from the 5×5 arrangement

(iii) Now, multiply remaining numbers row-wise, but write the resultants in colum-wise manner to directly get the adjoint.

Eg: $A=\begin{bmatrix}0&0&1\\2&-1&3\\1&1&4\end{bmatrix}$

For adj(A)



$$adj(A) = \begin{bmatrix}-7&1&1\\-5&-1&2\\3&0&0\end{bmatrix} \Rightarrow \boxed{A^{-1} = \frac{1}{3}\begin{bmatrix}-7&1&1\\-5&-1&2\\3&0&0\end{bmatrix}}$$ Ans

Properties of Inverse :->

1) $\boxed{AA^{-1} = A^{-1}A = I}$

2) $\boxed{|A^{-1}| = \frac{1}{|A|}}$

3) $\boxed{(AB)^{-1} = B^{-1}.A^{-1}}$

4) $\boxed{(kA)^{-1} = \frac{1}{k}.A^{-1}}$

5) $\boxed{(A^k)^{-1} = (A^{-1})^k = A^{-k}}$

6) $\boxed{(A^T)^{-1} = (A^{-1})^T}$

7) $\boxed{(A^{-1})^{-1} = A}$

Ques If $A=\begin{bmatrix}2&-1&1\\-1&2&-1\\1&-1&2\end{bmatrix}$ then find the value of $|3adj(2A^{-1})|$.

Sol^n

$$M = |3\,adj(2A^{-1})|$$

$$M = 3^3\,|adj(2A^{-1})| \Rightarrow 3^3\,|2A^{-1}|^2$$

$$M = 3^3\,2^6\,|A^{-1}|^2 \Rightarrow \frac{6^3.2^3}{|A|^2}$$

Now, $|A| = 2(4-1)+1(-2+1)+1(1-2)$

$= 6-1-1 \Rightarrow 2^2$

So, $M = \frac{6^3.2^3}{2^4} \Rightarrow \frac{6^3}{2} \Rightarrow \boxed{M = 108}$ Ans

Ques If $A=\begin{bmatrix}0&-\tan(\theta_2)\\ \tan(\theta_2)&0\end{bmatrix}$ and $(I_2+A)(I_2-A)^{-1} = \begin{bmatrix}a&-b\\b&a\end{bmatrix}$, then $13(a^2+b^2)$ is equal to ____.

Sol^n Since values of a & b are independent of θ, we assume $(\theta = 0^\circ)$

$A=\begin{bmatrix}0&0\\0&0\end{bmatrix}$ | So, $(I_2+0)(I_2-0)^{-1} = I_2.I_2^{-1} = I_2$

i.e. $\boxed{a=1}$ & $\boxed{b=0}$

So, $\boxed{13(a^2+b^2) = 13}$ Ans

Ques Let M & N be two 3×3 non-~~symm~~ singular skew symmetric matrices such that $MN=NM$, then find $M^2N^2(M^TN)^{-1}(MN^{-1})^T$.

Sol^n It is given, $\boxed{M^T=-M}$ & $\boxed{N^T=-N}$

$\boxed{MN=NM}$

Now,

$$= M^2N^2(M^TN)^{-1}(MN^{-1})^T$$

$$= M^2N.N.N^{-1}.(M^T)^{-1}(MN^{-1})^T$$

$$= M^2N(M^T)^{-1}(N^{-1})^T.M^T$$

$$= M(MN)(M^TN^T)^{-1}.M^T$$

$$= M(MN)(MN)^{-1}.M^T \quad [\because M^TN^T = (-M)(-N) = MN]$$

$$= M.M^T \Rightarrow M(-M)$$

$$= -M^2$$ Ans